

```
[root@abc ~]# nmcli con show "ens10" | grep -i ipv
ipv4.method:                manual
ipv4.dns:                   192.168.123.1
ipv4.dns-search:            example.com
ipv4.addresses:             192.168.123.13/23
ipv4.gateway:               192.168.123.1
ipv4.routes:
ipv4.route-metric:          -1
ipv4.ignore-auto-routes:    no
ipv4.ignore-auto-dns:       no
ipv4.dhcp-client-id:        --
ipv4.dhcp-send-hostname:    yes
ipv4.dhcp-hostname:         --
ipv4.never-default:         no
ipv4.may-fail:              yes
ipv6.method:                auto
ipv6.dns:
ipv6.dns-search:
ipv6.addresses:
ipv6.gateway:               --
ipv6.routes:
ipv6.route-metric:          -1
ipv6.ignore-auto-routes:    no
ipv6.ignore-auto-dns:       no
ipv6.never-default:         no
ipv6.may-fail:              yes
ipv6.ip6-privacy:           -1 (unknown)
ipv6.dhcp-send-hostname:    yes
ipv6.dhcp-hostname:         --
[root@abc ~]#
```

Figure 5: Ipv4 or IPv6 configuration

is shown in Figure 5.

The *ipv4.method* and *ipv6.method* settings define how IPv4 and IPv6 addressing is configured on the interface. A value that auto indicates DHCP is used for network configuration. When *manual* is specified for an IP protocol, then IP information must be specified with addresses and gateway settings. The network manager updates the network interface configuration files when *nmcli* is used to change and assign values to these settings. Another approach would be to change the configuration files in */etc/sysconfig/network-scripts*, and then use the command *nmcli con reload* to have the network manager read them for configuration changes.

The network manager's network interface configuration changes are not applied to an interface if it is already active. In this case, the interface must be bounced, brought down and back up, to effect the configuration changes. The following commands can be used to bring down the interface and then bring it up again:

```
# nmcli con down ens10
# nmcli con up ens10
```

The same has been shown in Figure 6 for device *ens10*.

Restarting affected network services

Network services can still be unavailable when network interface configuration changes have been made. The affected

```
[root@abc Desktop]# nmcli con down ens10
Connection 'ens10' successfully deactivated (D-Bus active path: /org/freedesktop/NetworkManager/ActiveConnection/3)
[root@abc Desktop]# nmcli con up ens10
Connection successfully activated (D-Bus active path: /org/freedesktop/NetworkManager/ActiveConnection/4)
[root@abc Desktop]#
```

Figure 6: The interface was first disabled and then enabled

```
[root@abc ~]# cat /etc/resolv.conf
# Generated by NetworkManager
search example.com
nameserver 192.168.123.1
[root@abc ~]#
```

Figure 7: */etc/resolv.conf* file

services should be restarted so they listen to the new address on a changed network interface. Use the following command to restart the network services:

```
# systemctl restart network
```

Solving DNS resolution problems

Client-side DNS resolution problems can often cause services to be sluggish or unavailable. For example, the SSH daemon has a UseDNS SSH setting. It determines whether *sshd* should check if the host name matches the IP address of incoming connections. This feature is enabled, by default, in Linux systems.

The main configuration file for DNS resolution is */etc/resolv.conf*. This file identifies the current DNS servers.

Figure 7 is a snapshot of */etc/resolv.conf*.

/etc/resolv.conf is a good starting point for DNS troubleshooting, this file is often managed by other scripts. The DNS variables defined in the network interface files, */etc/sysconfig/network-scripts/ifcfg-**, can define the name server entries in */etc/resolv.conf*. The PEERDNS variable must be set to 'Yes' for these settings to be applied. Figure 8 shows how network configuration settings and DNS values are assigned on this host.

Note: The *PEERDNS=yes* setting also configures *dhclient* to modify */etc/resolv.conf* when using DHCP to assign network values.

When the network manager is active, it makes changes to the network interface configuration files in */etc/sysconfig/network-scripts*. In these cases, DNS configuration changes need to be made to the network manager's configuration information. This is accomplished using the *nmcli* command. Use the following commands for the same:

```
# nmcli conn show <device-name> | grep dns
// To check the DNS.
# nmcli conn mod <device-name> ipv4.dns <dns-value>
// The modified DNS that you want to achieve.
# nmcli conn show <device-name> | grep dns
//To verify the DNS
```

```
[root@abc ~]# grep -i dns /etc/sysconfig/network-scripts/ifcfg-*
/etc/sysconfig/network-scripts/ifcfg-ens10:DNS1=192.168.123.1
/etc/sysconfig/network-scripts/ifcfg-ens10:IPV6_PEERDNS=yes
/etc/sysconfig/network-scripts/ifcfg-eth0:DNS1=192.168.123.1
/etc/sysconfig/network-scripts/ifcfg-eth0:IPV6_PEERDNS=yes
[root@abc ~]#
```

Figure 8: PEERDNS value is assigned as 'Yes'